Phase 2 – Glycolytic Reversal / Reduction

Phospho glyceric acid is phosphorylated by ATP and produces 1,3 bis phospho glyceric acid by PGA kinase. 1,3 bis phospho glyceric acid is reduced to glyceraldehyde 3 Phosphate (G-3-P) by using the reducing power NADPH + H^+. Glyceraldehyde 3 phosphate is converted into its isomeric form di hydroxy acetone phosphate (DHAP).

PGA $\xrightarrow[\text{ATP}]{\text{PGA Kinase}}$ 1,3 bisphosphoglyceric acid (ADP)

1,3 bisphosphoglceric acid $\xrightarrow{\text{NADPH + H}^+}$ Glyceraldehyde-3-Phosphate (NADP^+)

Phase 3 – Regeneration

Regeneration of RUBP involves the formation of several intermediate compounds of 6-carbon, 5-carbon,4-carbon and 7- carbon skeleton. Fixation of one carbon dioxide requires 3 ATPs and 2 NADPH + H^+, and for the fixation of 6 CO_2 requires 18 ATPs and 12 NADPH + H^+ during C_3 cycle. One 6 carbon compound is the net gain to form hexose sugar.

RU5P $\xrightarrow{\text{ATP}}$ RUBP (ADP)

Overall equation for dark reaction:

$$6CO_2 + 18ATP + 12NADPH + H^+ \longrightarrow C_6H_{12}O_6 + 6H_2O + 18ADP + 18Pi + 12NADP^+$$



RUBISCO – RUBP Carboxylase Oxygenase enzyme, is the most abundant protein found on earth. It constitutes 16 % of the chloroplast protein. It acts as carboxylase in the presence of CO_2 and oxygenase in the absence of CO_2.

13.12 Hatch & Slack Pathway or C_4 Cycle or Dicarboxylic Acid Pathway or Dicarboxylation Pathway

Till 1965, Calvin cycle is the only pathway for CO_2 fixation. But in 1965, **Kortschak**, **Hart** and **Burr** made observations in sugarcane and found C_4 or dicarboxylic acid pathway. Malate and aspartate are the major labelled products. This observation was confirmed by **Hatch** & **Slack** in 1967. This alternate pathway for the fixation of CO_2 was found in several tropical and sub-tropical grasses and some dicots. C_4 cycle is discovered in more than 1000 species. Among them 300 species belong to dicots and rest of them are monocots. C_4 plants represent about 5% of Earth's plant biomass and 1% of its known plant species. Despite this scarcity, they account for about 30% of terrestrial carbon fixation. Increasing the proportion of C_4 plants on earth could assist biosequestration of CO_2 and represent an important climate change avoidance strategy.

C_4 pathway is completed in two phases, first phase takes place in stroma of mesophyll cells, where the CO_2 acceptor molecule is 3-Carbon compound, phospho enol pyruvate (PEP) to form 4-carbon Oxalo acetic acid (OAA). The first product is a 4-carbon and so it is named as C_4 cycle. oxalo acetic acid is a dicarboxylic acid and hence this cycle is also known as **dicarboxylic acid pathway** (Figure 13.18). Carbon dioxide fixation takes place in two places one in mesophyll and another in bundle sheath cell (di carboxylation pathway). It is the adaptation of tropical and sub tropical plants growing in warm and dry conditions. Fixation of CO_2 with minimal loss is due to absence of photorespiration. C_4 plants require 5 ATP and 2 NADPH + H^+ to fix one molecule of CO_2.

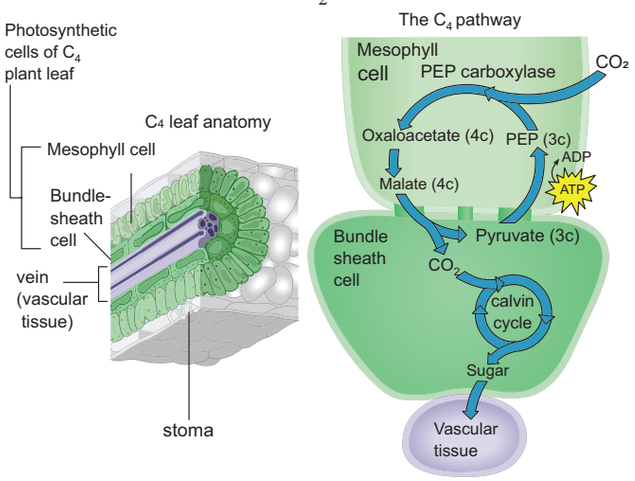


Figure 13.18: C_4 Cycle

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